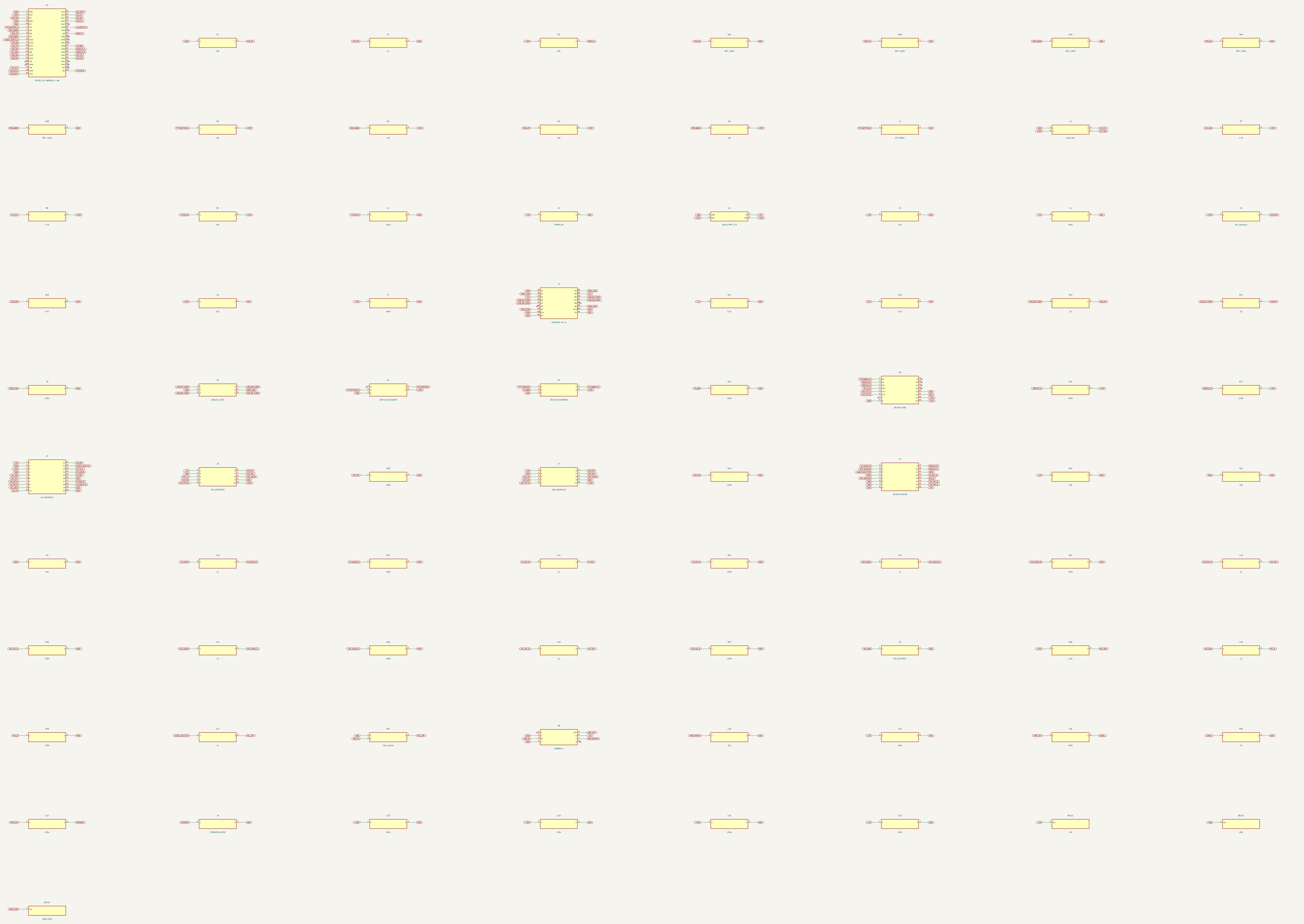
U1 ESP32-S3-WROOM-1-N8: GND 3V3 EN GND EP IO4 IO5 IO6 IO7 IO15 IO16 IO17 IO18 IO8 IO19 IO20 IO3 IO46 IO9 IO10 IO11 IO12 IO13 IO14 IO21 IO47 IO48 IO45 IO0 IO35 IO36 IO37 IO38 IO39 IO40 IO41 IO42 IO44 IO43 IO2 IO1
GND +3V3 ESP_EN GND GND PTT_BUTTON_N BTN_DOWN BTN_UP BTN_BAND CODEC_RESET_N VHF_RX VHF_TX UHF_RX I2C_SDA USB_DM USB_DP I2C_SCL I2S_MCLK I2S_BCLK
I2S_LRCK I2S_TX I2S_RX UHF_TX LO_RESET_N BOOT_N TX_ARM MODULE_A MODULE_B VHF_PD UHF_PD FILTER_ID
R1 +3V3 ESP_EN 10k
C1 ESP_EN GND 1u
R2 +3V3 BOOT_N 10k
SW1 ESP_EN GND B3F-1000
SW2 BOOT_N GND B3F-1000
SW3 BTN_DOWN GND B3F-1000
SW4 BTN_UP GND B3F-1000
SW5 BTN_BAND GND B3F-1000
R3 PTT_BUTTON_N +3V3 10k
R4 BTN_DOWN +3V3 10k
R5 BTN_UP +3V3 10k
R6 BTN_BAND +3V3 10k
J1 PTT_BUTTON_N GND PTT_PANEL
J2 GND +3V3 I2C_SCL I2C_SDA OLED_I2C
R7 I2C_SDA +3V3 4.7k
R8 I2C_SCL +3V3 4.7k
R9 FILTER_ID +3V3 10k
C2 FILTER_ID GND 100n
J3 +5V GND POWER_5V
U2 GND OUT IN TAB GND +3V3 +5V +3V3 LM1117IMP-3.3
C3 +5V GND 10u
C4 +5V GND 100n
C5 +3V3 LDO_ESR 22u tantalum
R10 LDO_ESR GND 0.47
C6 +3V3 GND 10u
C7 +3V3 GND 100n
J4 A1 A4 A5 A6 A7 A8 A9 A12 B1 B4 B5 B6 B7 B8 B9 B12 S1 GND VBUS_USB CC1 USB_DP_CONN USB_DM_CONN VBUS_USB GND GND VBUS_USB CC2 USB_DP_CONN USB_DM_CONN VBUS_USB GND GND USB4105-GF-A
R11 CC1 GND 5.1k
R12 CC2 GND 5.1k
R13 USB_DM_CONN USB_DM 22
R14 USB_DP_CONN USB_DP 22
C8 VBUS_USB GND 100n
U3 USB_DP_CONN GND USB_DM_CONN USB_DM_CONN VBUS_USB USB_DP_CONN USBLC6-2SC6
U4 A Y PTT_BUTTON_N GND PTT_PRESSED +3V3 SN74LVC1G04DBVR
U5 A B Y PTT_PRESSED TX_ARM GND TX_ENABLE_N +3V3 SN74LVC1G00DBVR
R15 TX_ARM GND 100k
U6 1G 1A 1B 1Y0 1Y1 1Y2 7 8 TX_ENABLE_N MODULE_A MODULE_B HF_TX_N VHF_PTT_N UHF_PTT_N GND GND GND +3V3 +3V3 SN74HC139D
R16 MODULE_A +3V3 100k
R17 MODULE_B +3V3 100k
J5 +5V GND +3V3 GND I2C_SDA I2C_SCL I2S_MCLK I2S_BCLK I2S_LRCK I2S_TX I2S_RX CODEC_RESET_N HF_TX_N HF_AUDIO HF_MIC GND FILTER_ID LO_RESET_N GND GND HF_INTERFACE
J6 +5V GND VHF_TX VHF_RX VHF_PTT_N VHF_PD VHF_MIC VHF_AUDIO GND +3V3 VHF_INTERFACE
R18 VHF_PD GND 100k
J7 +5V GND UHF_TX UHF_RX UHF_PTT_N UHF_PD UHF_MIC UHF_AUDIO GND +3V3 UHF_INTERFACE
R19 UHF_PD GND 100k
U7 HF_AUDIO_B UHF_AUDIO_B AUDIO_SELECTED VMID VHF_AUDIO_B GND GND GND MODULE_B MODULE_A VMID HF_MIC_B MIC_B VHF_MIC_B UHF_MIC_B +5V CD74HCT4052M
R20 +5V VMID 10k
R21 VMID GND 10k
C9 VMID GND 10u
C10 HF_AUDIO HF_AUDIO_B 1u
R22 HF_AUDIO_B VMID 100k
C11 HF_MIC_B HF_MIC 1u
R23 HF_MIC_B VMID 100k
C12 VHF_AUDIO VHF_AUDIO_B 1u
R24 VHF_AUDIO_B VMID 100k
C13 VHF_MIC_B VHF_MIC 1u
R25 VHF_MIC_B VMID 100k
C14 UHF_AUDIO UHF_AUDIO_B 1u
R26 UHF_AUDIO_B VMID 100k
C15 UHF_MIC_B UHF_MIC 1u
R27 UHF_MIC_B VMID 100k
J8 MIC_RAW GND MIC_ELECTRET
R28 +3V3 MIC_RAW 2.2k
C16 MIC_RAW MIC_B 1u
R29 MIC_B VMID 100k
C17 AUDIO_SELECTED VOL_TOP 1u
RV1 GND AMP_IN VOL_TOP 10k volume
U8 IN OUT GND AMP_IN GND AMP_OUT +5V AMP_BYPASS LM386N-1
C18 AMP_BYPASS GND 10u
C19 +5V GND 100n
C20 AMP_OUT ZOBEL 100n
R30 ZOBEL GND 10
C21 AMP_OUT SPEAKER 220u
J9 SPEAKER GND SPEAKER_8_OHM
C22 +3V3 GND 100n
C23 +3V3 GND 100n
C24 +3V3 GND 100n
C25 +5V GND 100n
#FLG1 pwr +5V +5V
#FLG2 pwr GND GND
#FLG3 pwr VBUS_USB VBUS_USB

Engineering prototype — not production released
Daniel Marques / SkyWave SDR
Sheet: /
File: control.kicad_sch
Title: SkyWave D1 — control
Size: A0
Date: 2026-10-12
Rev: D1-prototype
KiCad E.D.A. 10.0.7
Id: 1/1